

REAL - TIME CO-SEISMIC IONOSPHERIC DETECTOR (RCID): HAZARD ASSESSMENT IN NEAR - REAL - TIME



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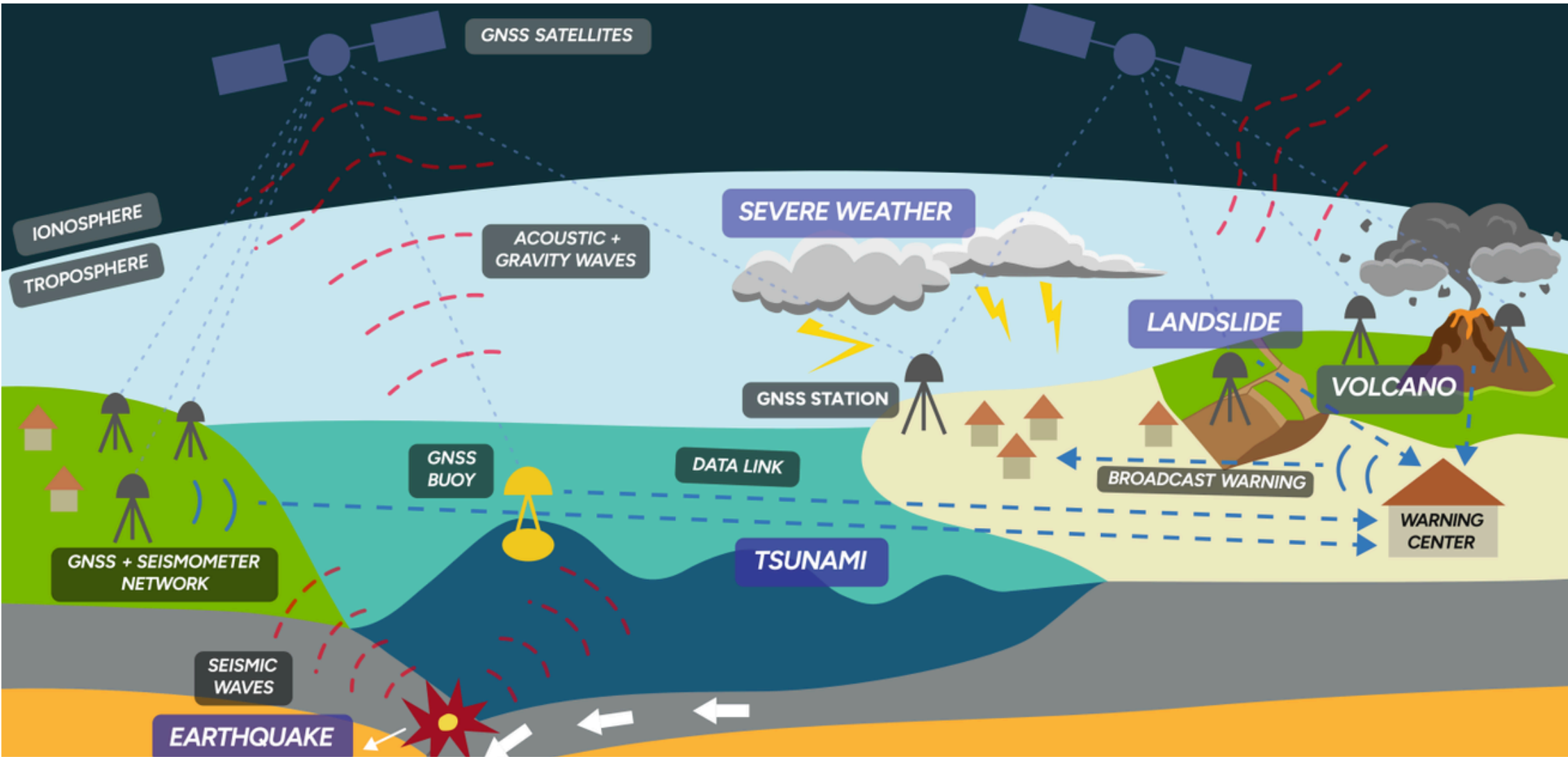
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PROBLEM STATEMENT



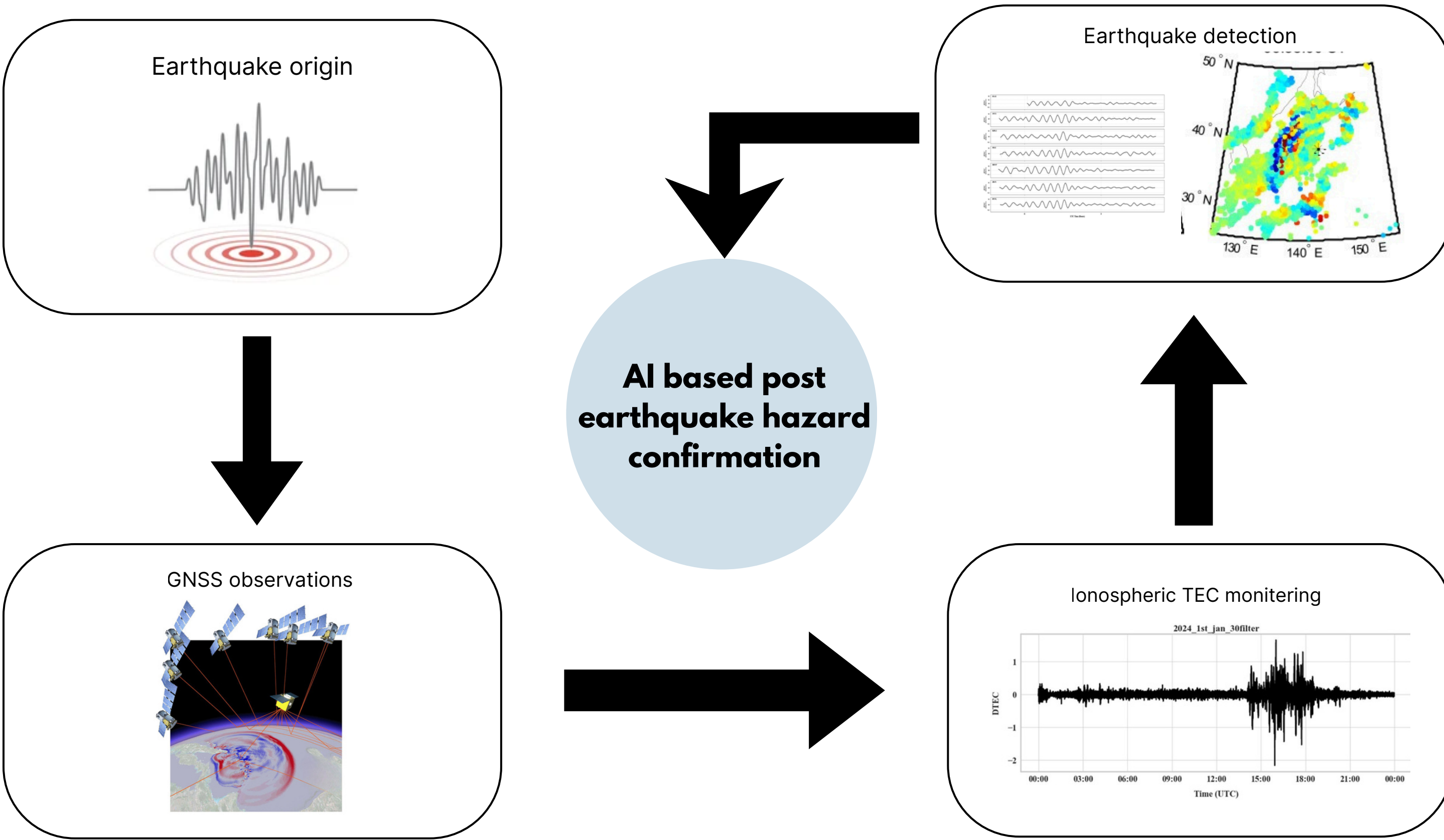
- **earthquake** any sudden shaking of the ground caused by the passage of seismic waves through Earth's rocks.
- **Impact:** Can cause ground shaking, landslides, tsunamis, and significant damage to property and infrastructure.
- Seismic waves from earthquakes trigger atmospheric acoustic/gravity waves that travel upward, causing detectable disturbances in the ionosphere's electron density.

- **Ionospheric Disturbance (TID):** The arriving waves disrupt the plasma (ionized particles) in the ionosphere, altering the number of electrons along paths between satellites and ground receivers (like GPS).

- Detection: These changes are measured as perturbations in the Total Electron Content (TEC), allowing scientists to observe the earthquake's effects from space.

PROPOSAL

- GNSS (Global Navigation Satellite System) can detect earthquake-induced acoustic and gravity waves by observing perturbations in the atmosphere and ionosphere. To overcome the limitations of conventional early warning systems in the timely identification of tsunami-generating earthquakes, particularly for near-field coastal regions.
- Our goal is to introduce and deploy the Real-Time Co-Seismic Ionospheric Detector (RCID), an AI-enabled framework that uses space-based sensing to provide rapid post-earthquake hazard confirmation.



EARTHQUAKE DETECTION METHODOLOGY

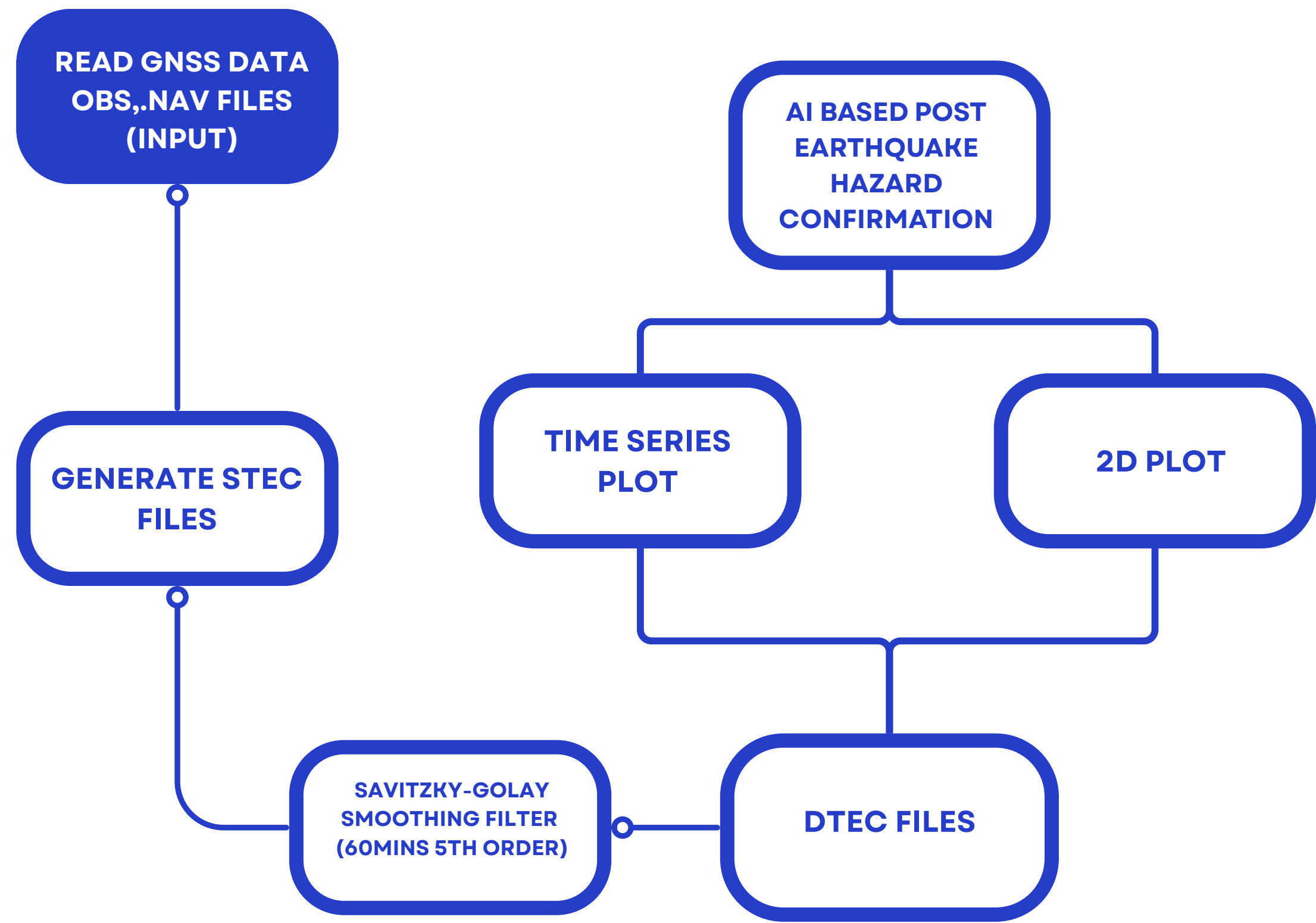


Figure: Observation methodology

High-Rate Data Acquisition

- **Source:** Multi-constellation GNSS data (GPS, GLONASS, Galileo, BDS).
- **Frequency:** Sampled at ≥ 1 Hz to capture transient acoustic signals.

Core Processing (IonKit-NH)

- **Calculation:** MATLAB-based toolkit performs dual-frequency calculations to extract Total Electron Content (TEC) and isolate anomalies.
- **Roadmap:** Future transition to RTCM 3.3 streams via NTRIP for in-memory, real-time decoding.

Feature Extraction

- **Time-Distance Maps:** Visualizes TEC vs. time/distance; slope indicates horizontal wave velocity (λ).
- **2D TEC Variation Maps:** Geographical snapshots revealing the concentric "N-type" waveform pattern.

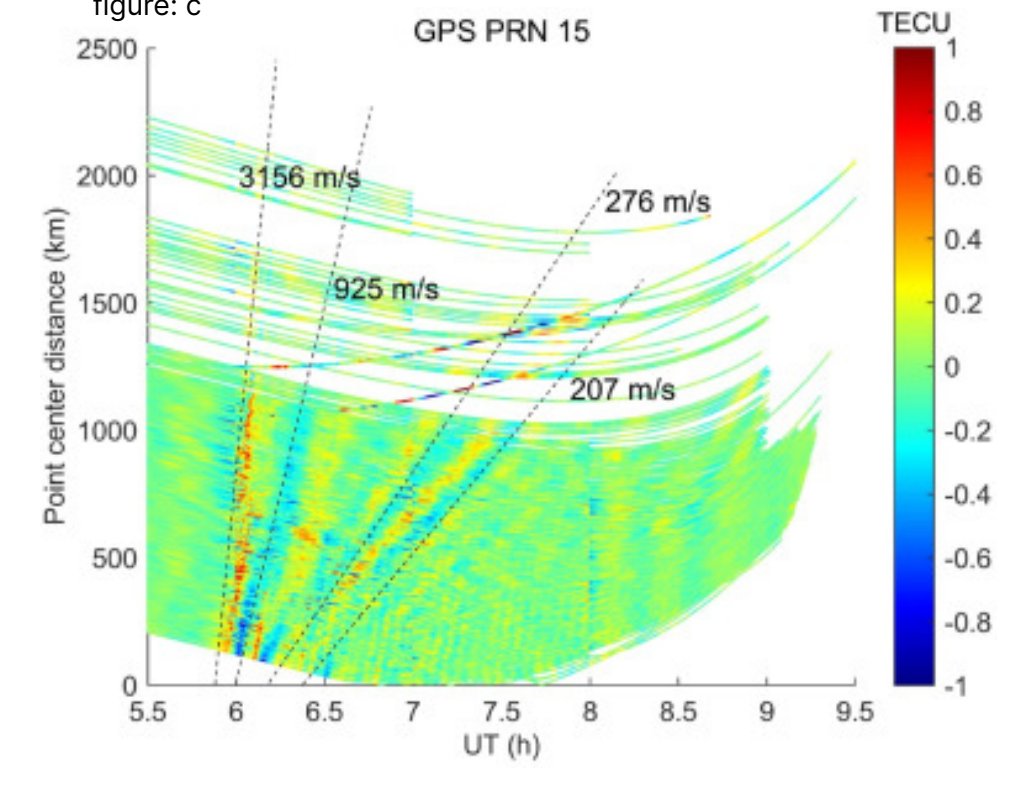
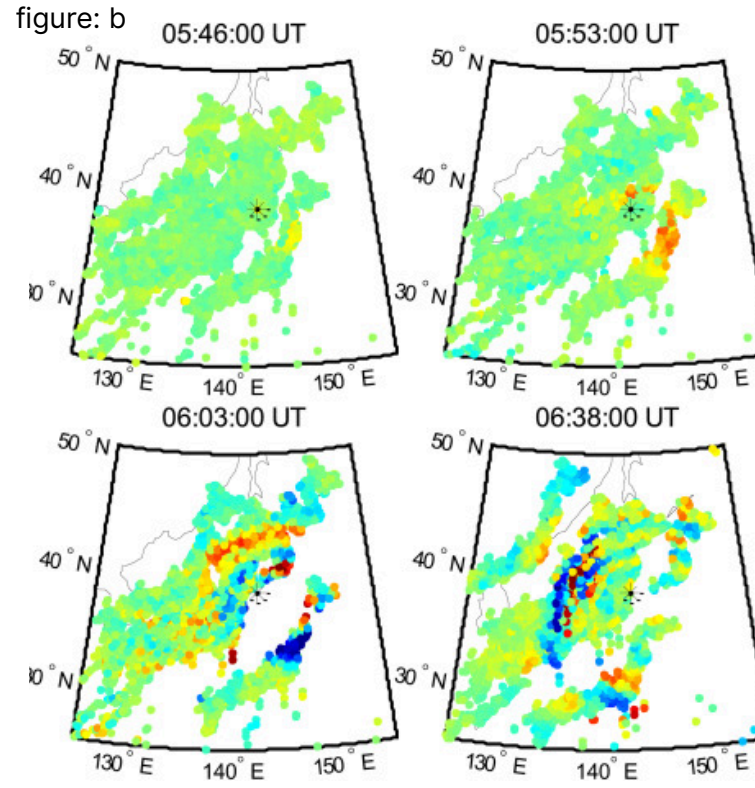
AI Pattern Recognition

- **CNN Architecture:** A Convolutional Neural Network processes 2D TEC maps as images.
- **Classification:** Instantaneously validates the "N-type" signature to confirm Co-Seismic Ionospheric Disturbances (CID).

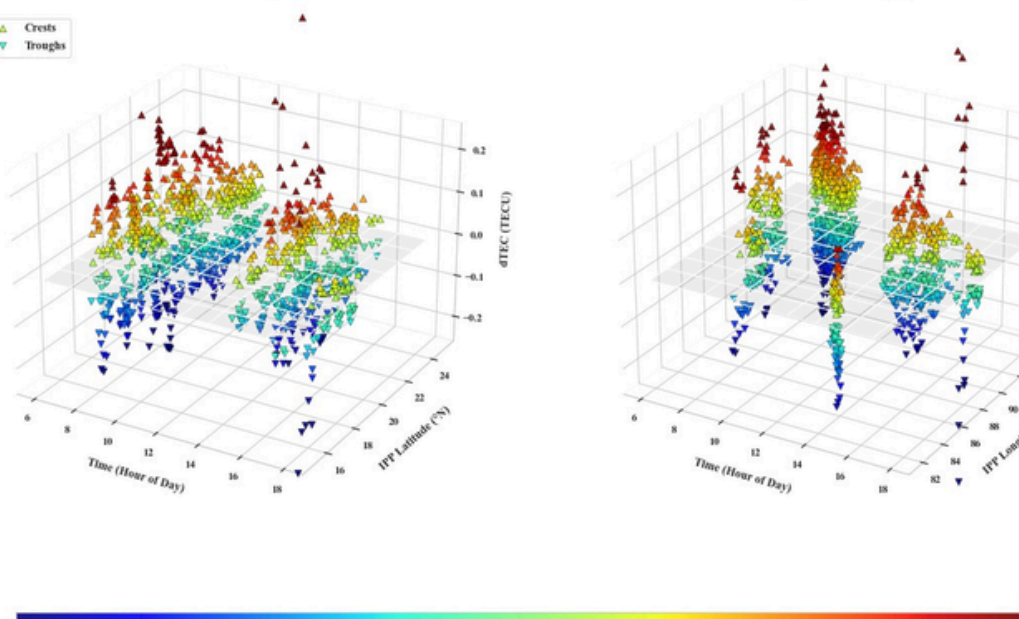
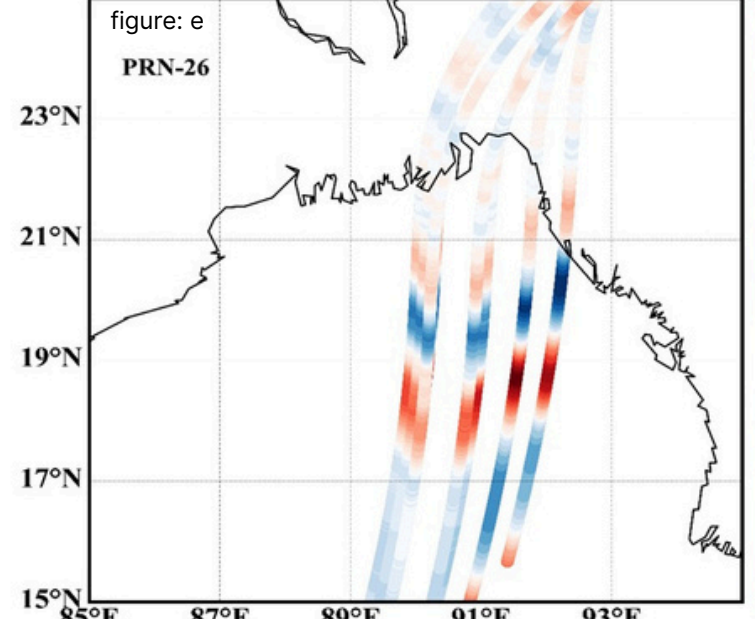
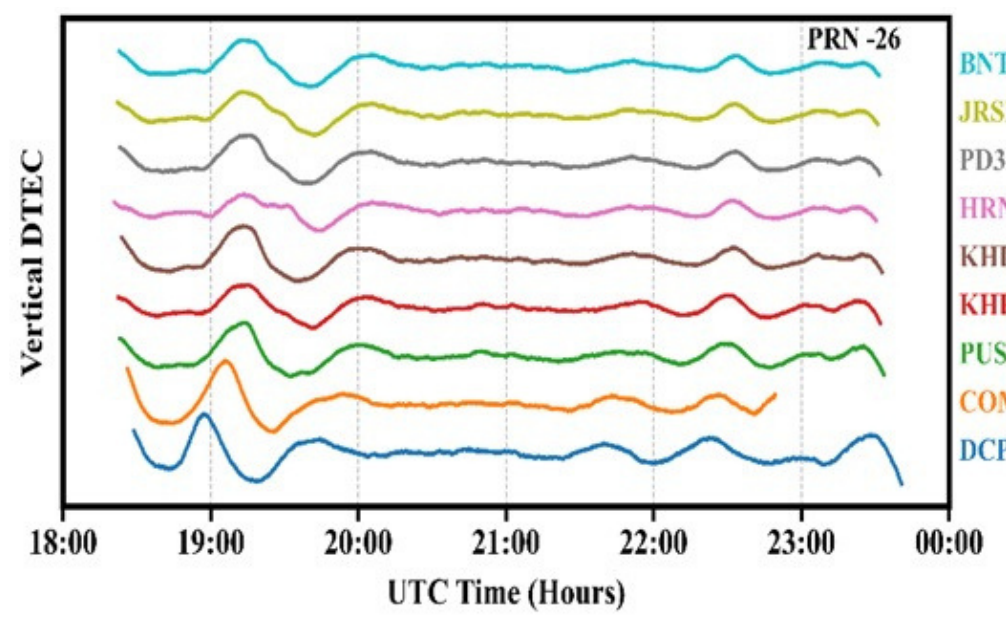
RESULTS



Case study of Japan earthquake (March 11, 2011)



Earthquake detection over indian region in future



DISCUSSIONS

- In this case study, the March 11, 2011 Japan earthquake was selected to implement the detection mechanism.
- Approximately 1,200 GNSS stations providing 30-s sampled data on the event day were collected from the GEONET network.
- There are no visible TIDs at and before the earthquake. About 7 min after the earthquake, namely at 05:53 UT, the TIDs begin to appear.
- At 06:03 UT, there are concentric circular TIDs with wide wavelength spread outward along the epicenter and band TIDs propagated in a southwesterly direction in ionosphere.
- In addition, at 06:38 UT, there are other concentric circular TIDs with narrow wavelength spread outward along the epicenter.
- As shown in Fig. 4, the propagation velocity of the observed TIDs varies from 207 m/s to 3,156 m/s. Specifically, the first-appearing band TIDs have a propagation velocity of about 3,000 m/s. The wide-wavelength concentric circular TIDs propagate at approximately 900 m/s, while the narrow-wavelength concentric circular TIDs exhibit different propagation velocities of about 200 m/s and 280 m/s. In addition, the initial locations of the narrow-wavelength concentric circular TIDs are approximately 500 km from the epicenter.

FUTURE SCOPE

Our goal is to introduce and deploy the Real-Time Co-Seismic Ionospheric Detector (RCID), an AI-enabled framework that uses space-based sensing to provide rapid post-earthquake hazard confirmation over India.

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